

The empirical recurrence for OEIS A236870: recovering a conjecture that is not written down

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Abstract

OEIS A236870 records “Empirical recurrence of order 96”, with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on a finite digraph and satisfies some monic recurrence of order at most the number of its states with distinct futures, here $S' = 194$; Berlekamp–Massey applied to $2S'$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 96, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 96 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry’s recurrence is the one displayed here, whatever its file contains, and it is proved.

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1 A conjecture one cannot read

OEIS A236870 is “Number of $(n+1) \times (3+1)$ 0..2 arrays with the maximum plus the upper median minus the lower median of every 2×2 subblock equal.”. It has offset 1 and begins

1069, 8117, 61750, 499947, 4012441, 32940090, 268846043, 2216853503, ...

The entry states:

Empirical recurrence of order 96 (see link above)

As of the “Last modified” line on the live entry (Jul 23 2025 09:37:03, revision 9) this is still recorded as empirical. The recurrence’s coefficients are not in the entry text.

2 The count is a walk count

The entry’s condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u , and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 194$ vertices, $a(n) = \iota^\top M^{j(n)} \tau$ for a linear reindexing j fixed by the entry’s shape and checked against its published terms. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \dots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 194$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S = 388$ exact terms of the model returns order 96 — the order the entry states — with the integer coefficients

$$a(n) = \sum_{i=1}^{96} c_i a(n-i),$$

c_1	19	c_{25}	−11929587209451815850	c_{49}	−3728841139500694304497002777
c_2	179	c_{26}	−479311753414849674706	c_{50}	−5669366808446416644823628609
c_3	−5366	c_{27}	242363890328464570827	c_{51}	6447975419480613724704879024
c_4	−6069	c_{28}	4195185206650405538410	c_{52}	8742588260720148614538931347
c_5	693450	c_{29}	−2934171017645251633005	c_{53}	−9600247542538326675768958099
c_6	−1264696	c_{30}	−31353991653878047470691	c_{54}	−11676032227899387211668359180
c_7	−54524871	c_{31}	26660329045787043681579	c_{55}	12291021351092751058774244348
c_8	191832602	c_{32}	201058394404767351452521	c_{56}	13474862827994520928636401203
c_9	2924685962	c_{33}	−194412101381675457769897	c_{57}	−13505556658440022209157487500
c_{10}	−14038833585	c_{34}	−1110489563134351145071516	c_{58}	−13400187344736059180460909890
c_{11}	−113630371984	c_{35}	1172150834578325104967168	c_{59}	12704490713950766193145320586
c_{12}	681860222066	c_{36}	5299128713493462951680983	c_{60}	11444050142103585321610246209
c_{13}	3307909182537	c_{37}	−5938190978233033603223486	c_{61}	−10198504967620131990296720935
c_{14}	−24292660628095	c_{38}	−21899389319201902049359031	c_{62}	−8359369101473248576966645645
c_{15}	−73371877560757	c_{39}	25525879372541095304445464	c_{63}	6959098958719805065725486811
c_{16}	667570845627245	c_{40}	78518930600782646236031519	c_{64}	5197837159739211526269554292
c_{17}	1239608742783699	c_{41}	−93689878412462547949010759	c_{65}	−4017584950889768517169278837
c_{18}	−14593691152712584	c_{42}	−244550793217826382074474250	c_{66}	−2735962582454230710321264820
c_{19}	−15490068790141455	c_{43}	294836777353247894451861044	c_{67}	1951391083445293954247325989
c_{20}	259210984432489364	c_{44}	662108477889477389745682914	c_{68}	1211237551060113133260842929
c_{21}	127074381261634881	c_{45}	−797627857547491245305673901	c_{69}	−792178164216400210851110478
c_{22}	−3798991329550340787	c_{46}	−1558664914938474646724347645	c_{70}	−447647553001953017257666667
c_{23}	−250593646522391882	c_{47}	1857929079355027454490139487	c_{71}	266702415602707888000296470
c_{24}	46485935999979431626	c_{48}	3189540524441175594911924372	c_{72}	136929102081098854174625540

Theorem 3. *The recurrence above holds for every n , and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 96, possibly only from some index on. Let q be that recurrence’s characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $96 + 4$ terms removed returns order 96 as well, so $\deg q_{\min} = 96 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. $\square$

4 Verification

Three checks.

First, the walk model reproduces all 17 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $96 + 4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture’s statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 96 that this sequence satisfies — and that family turns out to have exactly one member.

References

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